SELECTION SORT PROGRAM

```
#include <stdio.h>
int smallest(int arr[], int k, int n)
{
int pos = k, small=arr[k], i;
for(i=k+1;i<n;i++)
{
if(arr[i]< small)
{
small = arr[i];
pos = i;
}
}
return pos;
}
void selection_sort(int arr[],int n)
{
int k, pos, temp;
for(k=0;k<n;k++)
{
pos = smallest(arr, k, n);
temp = arr[k];
arr[k] = arr[pos];
arr[pos] = temp;
}
}
void main() {
int arr[10], i, n;
printf("\n Enter the number of elements in the array: ");
scanf("%d", &n);
printf("\n Enter the elements of the array: ");
```

```
for(i=0;i<n;i++)
{
printf("\n enter the arr[%d] element: ",i+1);
scanf("%d", &arr[i]);
}
selection_sort(arr, n);
printf("\n The sorted array is: \n");
for(i=0;i<n;i++)
printf(" %d\t", arr[i]);
}
```

OUTPUT :

Enter the number of elements in the array: 5

Enter the elements of the array:

enter the arr[1] element: 3

enter the arr[2] element: 4

enter the arr[3] element: 5

enter the arr[4] element: 2

enter the arr[5] element: 1

The sorted array is: 1 2 3 4 5

ALGORITHM:

SMALLEST (ARR , K , N , POS)

Step 1: [INITIALIZE] SET SMALL = ARR[K]

Step 2: [INITIALIZE] SET POS = K

Step 3: Repeat for J = K+1 to N

IF SMALL > ARR[J]

SET SMALL = ARR[J]

SET POS = J

[END OF IF]

[END OF LOOP]

Step 4: RETURN POS

SELECTION SORT(ARR , N)

Step 1: repeat step 2 and 3 for k = 1 to N-1

Step 2: Call Smallest(ARR , K , N , POS)

Step 3: SWAP A[K] with ARR[POS]

[END OF LOOP]

Step 4: EXIT